PSSTFN: A Progressive Spatial–Temporal–Spectral Fusion Network for Remote Sensing Images

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***Abstract*— Spatial–temporal–spectral fusion (STSF) is highly desirable to generate dense-time image series with high spatial and spectral resolution by integrating the complementary advantages of multisource and multitemporal observations. However, most existing STSF methods are still limited to the assumption of linear temporal, spatial, and spectral relationships. In addition, the STSF methods on Landsat and MODIS data are insufficient to characterize the inherent properties of the current spaceborne hyperspectral (HS) images with low spatial and temporal resolutions. For these, we propose a progressive STSF network (PSSTFN) by interestingly integrating spatial–spectral fusion and spectral–temporal fusion into a unified end-to-end STSF framework. Specifically, in the spatial–spectral fusion stage, we obtain the hierarchical features with different receptive fields and propose a multiattention guided (MAG) module for joint learning and refinement of spatial–spectral features. In the spectral–temporal fusion stage, a feature insertion module (FIM) is presented to embed the difference images into the resulting spatial–spectral features, and the estimation from deeper layers is cascaded for more reliable spatial information. We build the Dongying (DY) and Yellow River Estuary (YRE) remote sensing datasets based on Sentinel-2 and ZiYuan (ZY)-1 02D satellites for verification, and the experimental results on reduce- and full-resolution data demonstrate the superior performance of our method over the existing methods visually and quantitatively.**

***Index Terms*— Deep learning (DL), Sentinel-2 multispectral (MS) image, spatial–temporal–spectral fusion (STSF), Ziyuan (ZY)-1 02D hyperspectral (HS) image.**

I. Introduction

REMOTE sensing images with all high temporal, high spatial, and high spectral resolutions are urgently needed in many fields, such as dynamic monitoring of land surface features, fine agriculture, disaster monitoring, and early warning [1], [2], [3]. However, limited by the hardware and other conditions of satellite sensor imaging systems, such as signal-to-noise ratio, data storage, and transmission [4], current satellite sensors have to make a fundamental tradeoff among the spectral, spatial, and temporal resolutions [5], as shown in Fig. 1. For example, Ziyuan (ZY)-1 02D satellite provides hyperspectral (HS) image with high spectral resolution (166 bands covering the wavelength range from 400 to 2500 nm), but the spatial resolution (30 m) and revisited period (55 days) are relatively lower. Sentinel-2 satellite provides multispectral (MS) image featured by a relatively higher spatial resolution (10 m) and revisited period (five days), but the spectral resolution (13 bands covering the wavelength range from 400 to 2400 nm) is insufficient [6]. Fortunately, multisource remote sensing image fusion is an effective way to alleviate the intrinsic limitations [7].

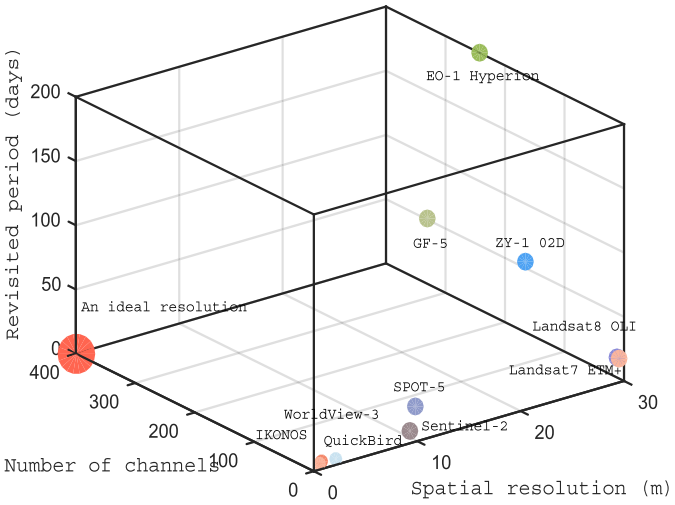


Fig. 1. Current satellites provide remote sensing images with a fundamental tradeoff between spatial, temporal, and spectral resolutions.

Image fusion algorithms for remote sensing have made remarkable strides and are divided into three main categories in application orientation: spatial–spectral fusion, spatial–temporal fusion, and spatial–temporal–spectral fusion (STSF) [8].

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A. Spatial–Spectral Fusion

Spatial–spectral fusion aims to overcome the tradeoff between the spatial and spectral resolutions of remote sensing observations. From the perspective of fusion objects, the existing spatial–spectral fusion methods mainly include panchromatic (PAN)/MS image fusion (i.e., pansharpening), PAN/HS or MS/HS image fusion, and PAN/MS/HS fusion [7], [9], [10], [11], [12]. As a core task, pansharpening sharpens the spatial textures of MS images drawing support from PAN images, and many fusion tasks are born out of pansharpening-based channelwise calculation or extensions of its connotation. From the perspective of fusion algorithms, the classical spatial–spectral fusion algorithms can be classified into the following categories: 1) component substitution (CS)-based methods [13], [14], [15]; 2) multiresolution analysis (MRA)-based methods [16], [17], [18]; 3) variational optimization (VO)-based methods [19], [20]; and 4) learning-based methods [21], [22]. Take pansharpening as an example, the CS-based method replaces the spatial components of low-resolution (LR)-MS image with the high-resolution (HR)-PAN image to achieve the spatial enhancement of the MS image, such as intensity–hue–saturation (IHS) [23]. The MRA-based method extracts spatial information through MRA tools, such as wavelet transform and Laplacian pyramid, and the extracted spatial details are then injected into the MS image to obtain the high spatial resolution version, such as modulation-transfer-function-generalized Laplacian pyramid (MTF-GLP) [24]. However, the above methods cannot make a good compromise in spatial and spectral fidelity. Most of the VO-based methods construct degeneration models to describe the relationships between the desired images and the observations. However, it generally needs relatively larger parameters to achieve an acceptable result [25]. The learning-based methods such as deep learning (DL) can better explore the concealed relationships among multisource data. Various deep-learning-based spatial–spectral fusion networks have been proposed and achieved better performance than the traditional methods. For example, the simple three-layer convolutional neural network (PNN) [26], a two-stream fusion network (TFNet) [27], and HyperTransformer [28].

B. Spatial–Temporal Fusion

Spatial–temporal fusion aims to blend the complementary temporal and spatial information of multisource images to acquire continuous observation series. From the perspective of fusion methods, the existing spatial–temporal fusion methods include [29]: 1) unmixing-based methods; 2) weight-function-based methods; 3) Bayesian-based methods; and 4) learning-based methods. The unmixing-based method is proposed on the linear mixing spectral model (LMM), which decomposes coarse pixels for fine pixel estimation, such as the multisensor multiresolution technique (MMT) [30]. The weight-function-based method predicts the fusion results by combining the complementary spatial and temporal information of the observations through a weight filtering function, such as spatial and temporal adaptive reflectance fusion model (STARFM) [31]. The Bayesian-based method establishes the fusion model on the spatial–temporal relationship functions through the Bayesian estimation theory [32], [33]. However, the traditional method generally performs mediocrely in complex land cover variations; compared with this, the learning-based method builds intrinsic spatial–temporal correlation between LR–HR image pairs, which simultaneously introduces image super-resolution technology into the model [34], such as DL. The deep-learning-based method is now attracting much attention due to its competitive performance, such as a two-stream convolutional neural network (StfNet) [35], a spatio-temporal fusion method using a recurrent neural network (STFRNN) [36], and a multistream fusion network (MSNet) [37].

C. Spatial–Temporal–Spectral Fusion

Both spatial–spectral fusion and spatial–temporal fusion put the purpose on integrating two indicators among the three resolutions, which are unable to meet the demands of time-intensive and high-quality remote sensing applications. Therefore, it is meaningful to develop an STSF technique to maximize complementary characteristics from multisource remote sensing images. To date, several STSF methods have been proposed, and the classic methods can be roughly divided into two categories: stepwise-based STSF and integrated STSF. For the former, Chen et al. [38] decomposed the STSF into multiple one-on-one fusion modes for step-by-step processing to synthesize multisensor characteristics. For the latter, Shen [39] first proposed a maximum a posteriori (MAP)-based integrated fusion framework and achieved competitive results using simulated images. Huang et al. [40] exploited the relationships between temporal changes and spatial–spectral fusion and implemented them in real-data experiments (e.g., Landsat and MODIS datasets). Shen et al. [4] designed a unified STSF framework to fuse images from an arbitrary number of optical sensors with different spatial, temporal, and spectral resolutions. Zhao et al. [41] proposed a flexible integrated sparsity model by considering the spatial–spectral redundancy of multisource images. Then Peng et al. [42] introduced tensor factorization to formalize the fusion problem into core tensor and dictionary estimation, and the sparsity priors in spatial, temporal, and spectral dimensions were used for regularization.

While a few STSF methods have been proposed, some stubborn problems still existed. First, the existing methods use linear mappings to describe the relationship between observed images and desired images [38], [39], [40], [41], [42], which is hard to capture the abrupt land cover change happening during the fusion period. To our best knowledge, Jiang et al. [43] interestingly designed an integrated framework for heterogeneous STSF based on DL. Liu et al. [44] proposed an unsupervised network for joint learning of spatial–spectral information, which achieved accurate prediction on complex land cover changes. Second, most of the methods are oriented toward MODIS and Landsat data, assuming that HS images have higher temporal resolution than MS images, which is different from the widely applied spaceborne HS images such as ZY-1 02D, Hyperion, and GaoFen-5 [45]. To solve these

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problems, we creatively propose a progressive STSF network (PSSTFN) to learn the nonlinear correlation among multisource data,and achieve an overall improvement in resolution. Moreover, we develop two datasets collected from ZY-1 02D and Sentinel-2 satellites for STSF that are more suitable for current application demands. The main contributions of this article are summarized as follows.

1) We propose PSSTFN, which interestingly integrates spatial–spectral fusion and spectral–temporal fusion into a unified end-to-end framework. Compared with previous methods, the model design orients toward the characteristics of spaceborne HS images with lower temporal and spatial resolution.
2) In a progressive manner, spatial–spectral quality enhancement is performed ahead of temporal change estimation to diminish the difficulty of STSF. A multiattention guided (MAG) module is served for three feature streams and achieves joint refinement of effective content by the mutual guidance of attention mechanism. The feature insertion module (FIM) is designed to combine the resulting spatial–spectral features with difference image by considering the continuity of spectral structure and information redundancy.
3) The multisource datasets from ZY-1 02D and Sentinel-2 satellites, termed as Dongying (DY) and Yellow River Estuary (YRE), are created for validation. The results of reduced- and full-resolution experiments prove that our method achieves the best results in both spectral fidelity and estimation of land cover changes.

The rest of this article is organized as follows. Section II describes the framework of PSSTFN. Section III presents and discusses the experimental results. The summary and prospect of our work are presented in Section IV.

II. Proposed Method

For ease of understanding, we have annotated some repeatedly mentioned symbols. Taking ZY-1 02D HS images and Sentinel-2 MS images as examples, LR-HS at the auxiliary time $H_1 \in \mathbb{R}^{h\times w\times B}$ and HR-MS at the auxiliary time and target time $M_1 \in \mathbb{R}^{H\times W\times b}$ and $M_2 \in \mathbb{R}^{H\times W\times b}$ served as input. H (or h) and W (or w) denote the spatial dimensions, and B (or b) denotes the spectral band number ($H > h$, $W > w$, $B > b$). In addition, the results of the spatial–spectral fusion stage and spectral–temporal fusion stage are denoted as $F_1 \in \mathbb{R}^{H\times W\times B}$ and $F_{21} \in \mathbb{R}^{H\times W\times B}$, respectively. HR-HS $X_1 \in \mathbb{R}^{H\times W\times B}$ and $X_2 \in \mathbb{R}^{H\times W\times B}$ served as references. The framework of PSSTFN shown in Fig. 2, including two stages of the spatial–spectral fusion and spectral–temporal fusion, can be abstracted as followings:

$$F_1 = f_1[(H_1, M_1); \theta_1] \tag{1}$$

$$F_{21} = f_2[(F_1, M_1, M_2); \theta_2] \tag{2}$$

$$F_2 = f_3[(F_1, F_{21}); \theta_3] \tag{3}$$

where f_1 and f_2 denote the two fusion stages, f_3 denotes the reconstruction, and θ_1, θ_2, and θ_3 are the nonlinear learnable parameters.

A. Spatial–Spectral Fusion Stage

In the spatial–spectral fusion stage, we adopt two parallel branches (MS and HS branches) with similar structures to extract multiscale spatial and spectral features. In the designed extractors, the dilated convolutions with overlapping dilation rates ($\times 1$, $\times 2$, $\times 3$) are attached to reinforce the understanding of surface features, which avoids the negative grid effect and irreversible information loss caused by downscaling. The process is depicted as follows:

$$F_h^1, F_h^2, \ldots, F_h^l = \mathrm{FEM}(H_1) \tag{4}$$

$$F_m^1, F_m^2, \ldots, F_m^l = \mathrm{FEM}(M_1) \tag{5}$$

where FEM indicates a feature extractor consisting of dilated convolution and a residual block. The multiscale features $F_h^1, F_h^2, \ldots, F_h^l$ and $F_m^1, F_m^2, \ldots, F_m^l$ are then integrated through a 1×1 convolution as the intermediate modality. The MAG module is proposed to refine the coarse features, which considers the strong correlation between MS and HS features and enhances the valuable content with the guidance of the intermediate modality.

As shown in Fig. 3, MAG consists of the parallel spatial attention module and spectral attention module for joint refinement of effective spatial–spectral content. It is assumed that HS and MS branches provided individual spectral and spatial knowledge, respectively. Taking the lth layer as an example, we merge F_m^1 with F_h^1 to produce the intermediate feature F_c^1 through a 1×1 convolution. F_m^1 and F_h^1 are jointly fed into the spectral attention module to produce the spectralwise attention map for the adjustment of band values in F_m^1. Similarly, we get a spatial attention map for spatial texture enhancements in F_h^1. Under the mutual guidance, the refined features suppress the redundant common information. The formulas are expressed as follows:

$$F_{\mathrm{CA}} = F_{\mathrm{spe}}\left[\left(F_c^l, F_m^l\right); \alpha\right] \odot F_m^l \tag{6}$$

$$F_{\mathrm{SA}} = F_{\mathrm{spa}}\left[\left(F_c^l, F_h^l\right); \beta\right] \odot F_h^l \tag{7}$$

where $\odot$ represents elementwise multiplication, F_{spe} and F_{spa} represent the spectral attention module and spatial attention module, respectively, and F_{CA} and F_{SA} represent the refined features. F_{CA}, F_{SA}, and F_c^l are further integrated to produce F_G, and cascaded semantic information from higher levels can improve fusion accuracy. If F_1^{l-1} is available, it is blended with F_G through a 3×3 convolution. The formulas are expressed as follows:

$$F_G = \mathrm{Conv}_{3\times 3}\left(F_{\mathrm{CA}}, F_{\mathrm{SA}}, F_c^l\right) \tag{8}$$

$$F_1^l = \mathrm{Conv}_{3\times 3}\left(F_G, F_1^{l-1}\right) \tag{9}$$

where $\mathrm{Conv}_{3\times 3}$ indicates the 3×3 convolution operation. We use a 1×1 convolution to compress the channels of F_1^l to B, which is equal to that of the reference. On the grounds of spectral preservation and parameter reduction, F_1 is derived from the skip connection between bicubic upsampled H_1 and F_1^1 as the spatial–spectral fusion result.

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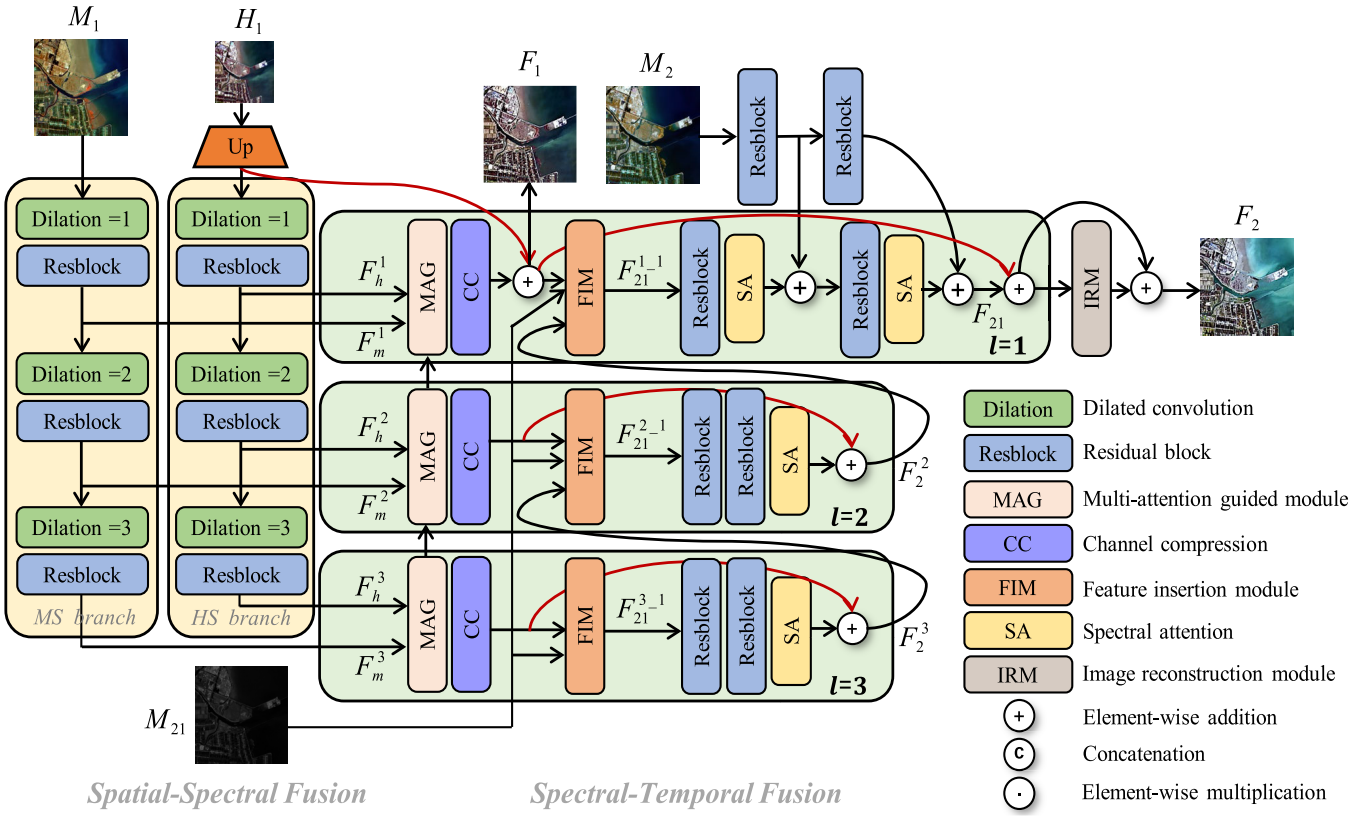


Fig. 2. Overview of PSSTFN.

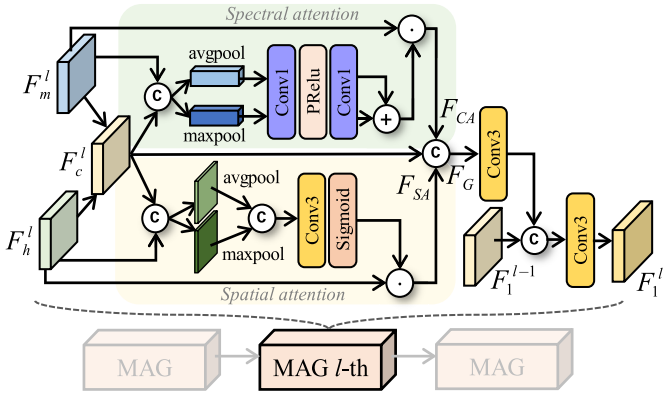


Fig. 3. Architecture of the MAG module.

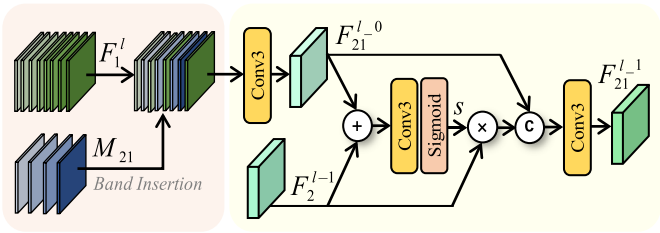


Fig. 4. Architecture of FIM.

B. Spectral–Temporal Fusion Stage

The spectral–temporal fusion stage predicts temporal change F_{21} with high spatial–spectral resolutions on the resulting spatial–spectral fusion.

MS images have a shorter revisited period, and therefore, the difference image M_{21} ($M_{21} = M_2 - M_1$) highlights the areas of change and is inserted into F_1^l to estimate the spectral content of the target time [33], which can be regarded as spectral super-resolution (SSR) with the assistance of HR-HS from auxiliary time.

To ensure a smooth transition at the intersection of the two stages, it is necessary to address the spectral fidelity caused by abrupt feature embedding and potential high-dimensional redundancy. For this, FIM is committed to maintaining HR-HS spectral structure and filtering redundant information. As shown in Fig. 4, FIM inserts the HR-MS image M_{21} into the HR-HS image F_1^l according to the spectral band relative position for the structural coherence.

If F_2^{l-1} is available, a gated unit is introduced to adaptively retain the information of interest with a learnable weight σ in F_2^{l-1}, thereby avoiding adverse common information. The process is as follows:

$$F_{21}^{l_0} = \mathrm{Conv}_{3\times3}\left[\mathrm{BI}\left(F_1^l, M_{21}\right)\right] \tag{10}$$

$$s = \sigma\left[\mathrm{Conv}_{3\times3}\left(F_{21}^{l_0} + F_2^{l-1}\right)\right] \tag{11}$$

$$F_{21}^{l_1} = \mathrm{Conv}_{1\times1}\left[\left(F_{21}^{l_0}, F_2^{l-1} * s\right)\right] \tag{12}$$

where BI indicates the band insertion operation, and four bands from HR-MS are inserted into HR-HS according to the wavelength range. $\mathrm{Conv}_{1\times1}$ indicates the 1×1 convolution operation, and σ indicates the Sigmoid function.

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There are slight differences between the bottom and other layers. First, the temporal change is mapped to the deep feature domain through residual blocks, and it generates F_{21}^l after spectral attention refinement. F_{21}^l is then added to the original spatial–spectral fusion feature F_1^l to characterize the rough STSF result F_2^l. Second, M_2 is injected into the bottom layer through residual blocks as supplements. The spectral attention module gets invoked after each convolution operation to suppress the redundant feature.

After gaining F_{21}, STSF result F_2 is transformed by an image reconstruction module (IRM) composed of three 3×3 convolution blocks as follows:

$$F_2 = \mathrm{Res}_i(F_1 + F_{21}) \tag{13}$$

where Res_i indicates the residual block with 3×3 size, $i = \{1, 2, 3\}$. F_2 is the desired image.

C. Loss Function

We design a set of weighted loss functions to constrain the network optimization, including mean absolute error (MAE) losses L_{MAE1} and L_{MAE2}, spatial loss L_{spa}, and spectral loss L_{spe}. The total loss function L is defined as follows:

$$L = \lambda_1 L_{\mathrm{MAE1}} + \lambda_2 L_{\mathrm{MAE2}} + \lambda_3 L_{\mathrm{spa}} + \lambda_4 L_{\mathrm{spe}} \tag{14}$$

where λ_1, λ_2, λ_3, and λ_4 denote the weights of each constraint term.

1) MAE Loss: MAE loss is the mean of the absolute value of the difference between the fusion image X_2 (or X_1) and the ground truth F_2 (or F_1). The MAE losses L_{MAE1} and L_{MAE2} are defined as

$$L_{\mathrm{MAE1}}(X_1, F_1) = \|X_1 - F_1\|_1 \tag{15}$$

$$L_{\mathrm{MAE2}}(X_2, F_2) = \|X_2 - F_2\|_1 \tag{16}$$

where $\|\cdot\|_1$ denotes the l_1-norm.

2) Spatial Loss: To reconstruct the image edge details better, we introduce spatial loss to retain more high-frequency information. Spatial loss is defined as

$$L_{\mathrm{spa}}(M_2, F_2) = \|\nabla_x M_2 - \nabla_x F_2\|_1 + \left\|\nabla_y M_2 - \nabla_y F_2\right\|_1 \tag{17}$$

where $\nabla_x M_2(h, w) = M_2(h, w) - M_2(h-1, w)$ denotes the gradient at pixel (h, w) in the horizontal direction. Similarly, $\nabla_y M_2(h, w) = M_2(h, w) - M_2(h-1, w)$ denotes the gradient at pixel (h, w) in the vertical direction. We use the sum of the two as the constraint of spatial details.

3) Spectral Loss: Spectral loss is important for channelwise HS information reconstruction. Spectral loss is defined as

$$L_{\mathrm{spe}}(X_2, F_2) = \frac{1}{HW} \sum_{h,w} \arccos\left(\frac{\left\langle \hat{X}_2^{h,w}, \hat{F}_2^{h,w} \right\rangle}{\left\| \hat{X}_2^{h,w} \right\|_2 \left\| \hat{F}_2^{h,w} \right\|_2} \right) \tag{18}$$

where H and W represent the number of pixels in rows and columns, respectively. $\langle \cdot, \cdot \rangle$ denotes the vector inner product. $\hat{X}_2^{h,w}$ and $\hat{F}_2^{h,w}$ represent the spectral vector at the pixel (h, w).

According to summarized experience, we set λ_1 as 1, λ_2 as 0.2, and λ_3 as 0.1. During the experiments, we assumed that the selection of λ_4 has a nonnegligible influence on the result [46], and therefore, the specific value of λ_4 will be discussed in the ablation experiment.

III. Experimental Results and Analysis

A. Dataset

The DY and YRE datasets collected from multisource and multitemporal Sentinel-2 HR-MS and ZY-1 02D LR-HS images are established for STSF to verify the capability of the proposed method. The details of the datasets are shown in Table I.

For the DY dataset, February 10, 2021 is nominated as the target time for prediction and December 11, 2019 is the auxiliary time. The task is to predict HR-HS image on February 10, 2021 with ZY-1 02D LR-HS image on December 11, 2019 and Sentinel-2 HR-MS images on December 8, 2019 and February 10, 2021. Due to the lack of real HR-HS image at February 10, 2021, the ZY-1 02D LR-HS image on February 10, 2021 is used as the reference. Given that there is only a three-day interval, we regard ZY-1 02D LR-HS image on December 11, 2019 and Sentinel-2 HR-MS images on December 8, 2019 as approximately matched. The data attributes of YRE are consistent with those of DY, but the former is a coastal geographical environment, while the latter covers a large range of farmland and buildings. Both the datasets contain a variety of landscapes, where distinct temporal changes have taken place between the two acquisition times, including seasonal spectral and land cover type variations. STSF works to obtain high spatial and spectral resolution image on target time, while eliminating the adverse effects of temporal changes.

In data processing, we dealt with the raw images through atmospheric correction, radiometric calibration, and image registration, then 15 noisy spectral bands were removed and 151 bands were retained from LR-HS images, and only the spectral bands with 10-m spatial resolutions in HR-MS images were kept as experimental data. Under Wald's protocol [47], the original images are blurred by a 7×7 Gaussian filter with a variance of 5 and decimated by the factor of 3, and the original LR-HS images are taken as references. We cropped the downsampling HR-MS and LR-HS images into $72 \times 72 \times 4$ and $24 \times 24 \times 151$ image patches without overlapping, and consequentially obtained 4000 pairs of image patches for the DY datasets and 1500 patches for the YRE datasets as the training data and the remaining images are reserved as testing data.

B. Quantitative Metrics

To perform the objective fusion quality assessment for the simulated data, we adopt five common metrics for quantitative assessment, including peak signal-to-noise ratio (PSNR), universal image quality index (UIQI), erreur relative globale adimensionnelle de synthèse (ERGAS), spectral angle mapper (SAM), and cross correlation (CC). PSNR, UIQI, and ERGAS are the metrics to measure the overall performance. SAM and CC measure the spectral and spatial qualities, respectively.

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TABLE I
SPECIFIC INFORMATION OF THE DY DATASET AND YRE DATASET

Dataset	Settings	Sentinel-2 (MS)	ZY1-02D (HS)
DY	Auxiliary time	December 8, 2019	December 11, 2019
	Target time	Feburary 10, 2021	Feburary 10, 2021
	Spatial resolution (m)	10	30
	Number of bands	4	151
	Wavelength (μm)	0.4-2.4	0.4-2.5
YRE	Auxiliary time	December 8, 2019	December 11, 2019
	Target time	Feburary 10, 2021	Feburary 10, 2021
	Spatial resolution (m)	10	30
	Number of bands	4	151
	Wavelength (μm)	0.4-2.4	0.4-2.5

1) Peak Signal-to-Noise Ratio: PSNR is used to represent the signal reconstruction quality of fused image in each band, and the average value is served as an overall assessment. The formula is expressed as

$$\mathrm{PSNR}\left(X_2^b, F_2^b\right) = 20\log_{10}\frac{\max\left(X_2^b\right)}{\sqrt{\mathrm{MSE}\left(X_2^b, F_2^b\right)}} \tag{19}$$

$$\mathrm{PSNR}(F_2, X_2) = \frac{1}{B}\sum_{b=1}^{B}\mathrm{PSNR}\left(X_2^b, F_2^b\right) \tag{20}$$

where F_2^b and X_2^b represent the bth band of the F_2 and ground truth X_2, respectively. MSE indicates mean square error function, and B represents the total number of bands. The ideal value of PSNR is $+\infty$.

2) Universal Image Quality Index: UIQI represents the degree of correlation, luminance distortion, and contrast distortion, which more conforms to human perception, and the formula is as follows:

$$\mathrm{UIQI}(F_2, X_2) = \frac{1}{HW}\sum_{h,w}\frac{4\sigma_{X,F}\bar{X}_2^{h,w}\bar{F}_2^{h,w}}{\left[\left(\bar{X}_2^{h,w}\right)^2 + \left(\bar{F}_2^{h,w}\right)^2\right]} \tag{21}$$

where $\bar{F}_2^{h,w}$ and $\bar{X}_2^{h,w}$ represent the mean values of F_2 and X_2 at pixel (h, w), respectively. $\sigma_{X,F}$ denotes the covariance. The ideal value of UIQI is 1.

3) Spectral Angle Mapper: SAM reflects the spectral similarity between F_2 and X_2, expressed as

$$\mathrm{SAM}(F_2, X_2) = \frac{1}{HW}\sum_{\mathrm{i},j}\arccos\left(\frac{\left\langle \hat{X}_2^{i,j}, \hat{F}_2^{i,j}\right\rangle}{\left\|\hat{X}_2^{i,j}\right\|_2\left\|\hat{F}_2^{i,j}\right\|_2}\right) \tag{22}$$

where H and W represent the number of pixels in rows and columns, respectively. $\hat{F}_2^{i,j}$ and $\hat{X}_2^{i,j}$ represent the spectral vector of F_2 and X_2 at the pixel (i, j), respectively. $\langle\cdot,\cdot\rangle$ denotes the vector inner product. The ideal value of SAM is 0.

4) Erreur Relative Globale Adimensionnelle de Synthèse: ERGAS comprehensively reflects the spatial and spectral qualities of F_2. The formula is as follows:

$$\mathrm{ERGAS}(F_2, X_2) = 100\frac{h}{o}\sqrt{\sum_{b=1}^{B}\left(\frac{\mathrm{RMSE}_b}{\left(\bar{X}_2^{h,w}\right)^2}\right)^2} \tag{23}$$

where h/o refers to the scale ratio of LR-HS and HR-MS images, and here, h/o is equal to 3. RMSE_b represents the RMSE of the bth band. The ideal value of ERGAS is 0.

5) Cross Correlation: CC is used to measure the spatial performance by averaging the CC value of each band. It is defined as

$$\mathrm{CC} = \frac{1}{B}\sum_{b=1}^{B}\frac{\sum_{h,w}\left[\left(F_2^b - \bar{F}_2^b\right)\left(X_2^b - \bar{X}_2^b\right)\right]}{\sqrt{\sum_{h,w}\left(F_2^b - \bar{F}_2^b\right)^2\sum_{h,w}\left(X_2^b - \bar{X}_2^b\right)^2}} \tag{24}$$

and the ideal value of CC is 1.

C. Implementation Details

To the best of our knowledge, the existing STSF methods are mainly founded on the hypothesis that the LR-HS images are featured with a high temporal resolution, such as the fusion of the Landsat-MODIS data, and they are not suitable for the most widely applied spaceborne LR-HS featured with low spatial and temporal resolutions. Therefore, we select three common SSR methods and a DL-based STSF method for comparison, including Arad [48], A+ [49], J-SLoL [50], and USTSFM [44].

1) Arad: Arad and Ben-Shahar [48] used a set of HS images as training and established a super complete dictionary D_H containing m HS atoms through k-singular value decomposition (K-SVD). Then the dictionary atoms were projected into the low spectral resolution (LSR) space to generate D_L. The orthogonal matching pursuit (OMP) algorithm is used to linearly decompose each pixel in the LSR image to calculate decomposition coefficient W, and then D_H and W are used to approximately reconstruct the HS images.

2) A+: Unlike Arad, A+ uses both the MS and HS atoms to calculate the projection matrix between the HSR space and LSR space. These projections are normalized and used to extract the c nearest neighbors of the training signatures for each dictionary atom [49]. After calculating the embedding matrices, the nearest dictionary atom is selected for each MS image pixel and A+ multiplies it with the corresponding stored projection matrix to get the HS image.

3) J-SLoL: J-SLoL used the common areas of HS images and MS images to learn low-rank overcomplete dictionaries and sparse coefficients [50]. J-SLoL further inferred the unknown HS images by sharing the sparse coefficients estimated using MS images.

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TABLE II
QUANTITATIVE ASSESSMENT OF COMPARISON METHODS ON THE DY DATASET. (THE BEST IS MARKED IN BOLD AND THE SECOND-BEST IS UNDERLINED)

Dataset	Method	PSNR	ERGAS	SAM	UIQI	CC
DY	Ideal value	$+\infty$	0	0	1	1
	Arad	25.2618	9.0028	9.9425	0.5791	0.8345
	A+	21.4922	9.4922	11.5861	0.8368	0.9089
	J-SLoL	23.8671	<u>5.9036</u>	7.2177	<u>0.9104</u>	**0.9209**
	USTSFM	<u>27.8910</u>	6.9481	<u>4.9622</u>	0.8796	0.8453
	proposed	**29.9937**	**5.2306**	**3.8021**	**0.9359**	<u>0.9167</u>

4) USTSFM: USTSFM implemented an effective unsupervised STSF network, which includes two jointly training task branches (namely, spatial–temporal–spectral mapping (STSM) and SSR). Compensation and optimization of dual-task in USTSFM promote spatial and spectral information learning between multisource images.

We implemented PSSTFN on NVIDIA RTX A6000 GPU based on the PyTorch framework. The parameters in our model were updated by Adam optimizer. We set the batchsize to 16 and the learning rate to 1×10^{-4} and added weight decay of 1×10^{-4} to prevent overfitting. We got the model with the best comprehensive performance near 1200 epochs.

D. Reduced-Resolution Experiments

The goal of STSF is to generate high spatial–spectral resolution HS images at the target time. In this experiment, the fusion results are visualized for intuitive analysis that we choose the RGB bands of the fused image (Red-band 29, Green-band 19, Blue-band 10) to synthesize the pseudocolor images for display.

Experimental Results on DY Dataset: Fig. 5 shows the fusion results produced by different methods on the DY dataset. We selected a region with the size of 756×180 for the test, which includes heterogeneous landscapes and land cover types that changed greatly between the two time phases. In all the comparison methods, Arad produces an image with very serious spatial blur while it is relatively good in spectral fidelity. The spatial details are enhanced excessively in J-SLoL and USTSFM, but they cannot settle spectral distortion. In contrast, PSSTFN achieves superior performance in spatial detail reconstruction and spectral preservation. It can be seen from Fig. 5(a) and (g) that seasonal alternation has produced significant differences in vegetation cover and waters, and some buildings suddenly appeared. Our proposed PSSTFN captures the temporal changes and reduces the errors caused by different phases while improving the spatial quality of the HS image.

In addition, to further evaluate the spectral fidelity among different methods [51], Fig. 6 shows the SAM map of each method on the DY dataset. The value of each pixel in the SAM map is between 0 and 1, and the closer the color is to the reference, the better the spectral fidelity is. Two areas worthy of attention are enclosed by red rectangles for better comparison. Arad and J-SLoL show large spectral distortions in lakes and ponds. Integral error of A+ is serious. It is obvious that the SAM map of PSSTFN produces the least spectral distortion than the DL-based USTSFM, and there is no abrupt serious distortion area in the relatively smooth error map.

TABLE III
SAM OF FULL-RESOLUTION EXPERIMENTAL RESULTS ON THE YRE DATASET. (THE BEST IS MARKED IN BOLD AND THE SECOND-BEST IS UNDERLINED)

Method	Arad	A+	J-SLoL	USTSFM	PSSTFN
SAM	4.9254	8.7506	5.4619	<u>4.0608</u>	**4.0550**

The quantitative assessment results are shown in Table II, and the best results are marked in bold. With regard to the DY dataset, for metric SAM, the behavior of PSSTFN is significantly better than the comparison methods, and for metric CC, PSSTFN is second only to J-SLoL. PSSTFN has great advantages in comprehensive metrics PSNR, ERGAS, and UIQI among all the methods. On the whole, PSSTFN generates the result with minimum losses, which demonstrates the competitive edge of PSSTFN in STSF ability.

Three kinds of typical landscapes are selected for comparison to prove the generalization ability of PSSTFN and visualize spectral reflectance details. We display the spectral reflectance of different landscapes, including waters, lands, and buildings, which is derived from the positions of specific pixels. As shown in Fig. 7, A+ performs larger spectral distortions in buildings and lands while J-SLoL is poor in waters. The DL-based methods exhibit a finer degree of curve-fitting, and PSSTFN outperforms in terms of effectiveness in different landscapes.

E. Full-Resolution Experiments

In this section, as the further supplements to the experiment of reduced-resolution, we conducted full-resolution experiments on real datasets. The original resolution HR-MS and LR-HS images are taken as the inputs of STSF.

Experimental Results on YRE Dataset: As shown in Fig. 8, comprehensive comparison with the original HR-MS and LR-HS image pairs illustrates that PSSTFN produces the image with better effect in spectral preservation and spatial detail injection and ensures the accurate prediction of temporal change. The urban area in the upper left corner can intuitively discover our advantages. As seen in the zoomed areas, the edge enhancement effect of Arad is not obvious, while A+, J-SLoL, and USTFM do not make full use of the LR-HS information. As for Fig. 9, the color of the SAM map of PSSTFN is darker than the comparison methods, especially in farmland and urban areas, which proves that PSSTFN is better in STSF. Due to the subtle differences in spectral fidelity between USTSFM and PSSTFN, the specific SAM values are shown in Table III for quantitative evaluation.

F. Ablation Experiment

In this section, we conduct a group of ablation experiments of PSSTFN on the DY dataset to further verify the effect of the designed components. The metric results are listed in Tables IV–VI.

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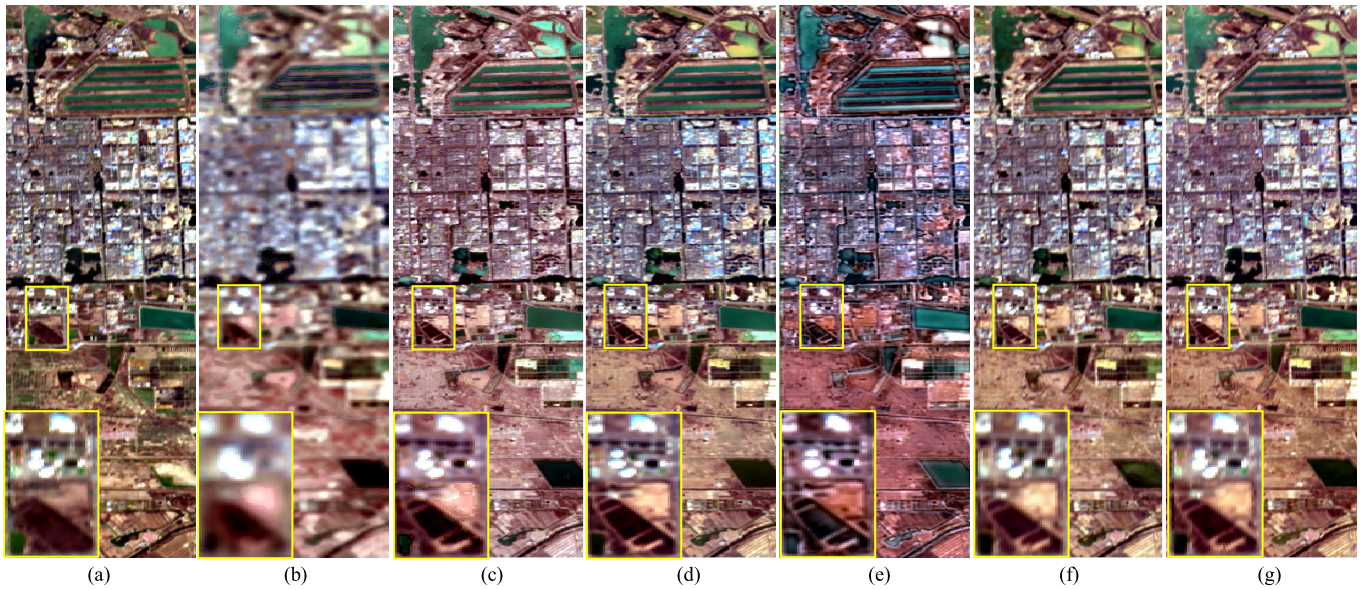


Fig. 5. Reduced-resolution experimental results on the DY dataset (displayed as pseudocolor). (a) HS image observed on December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) USTSFM. (f) PSSTFN. (g) HS image observed on February 10, 2021 (ground truth).

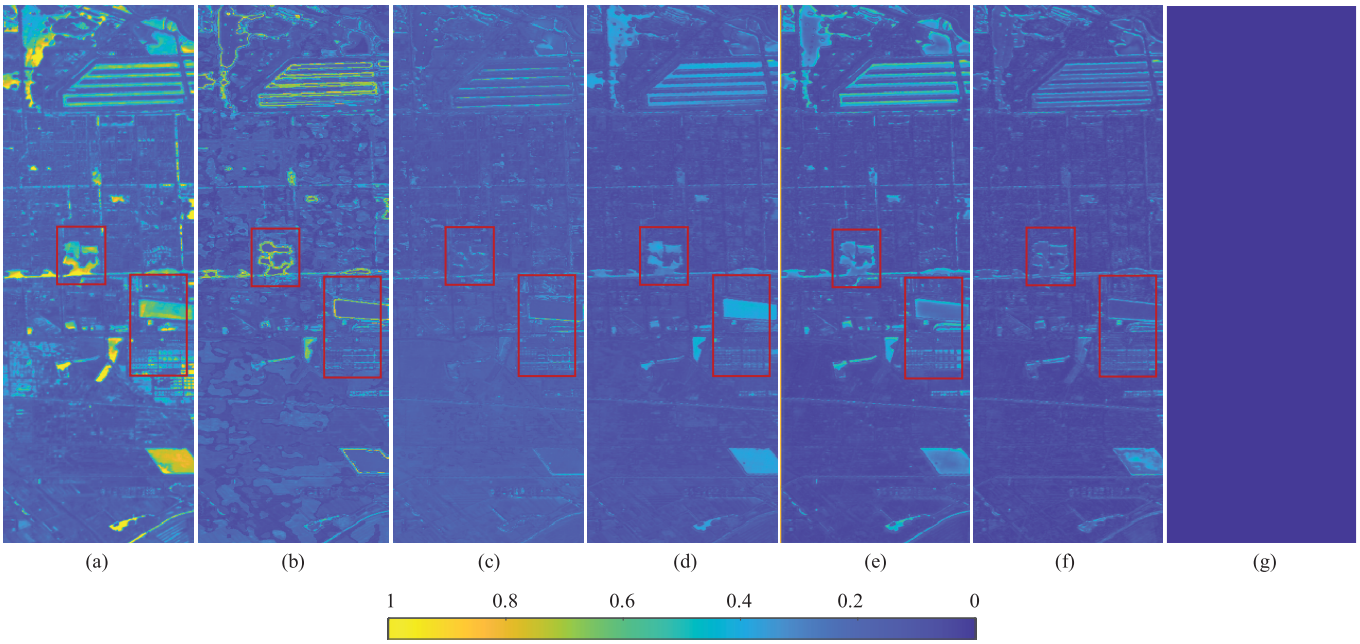


Fig. 6. SAM error maps of reduced-resolution experiments on the DY dataset. (a) HS image observed on December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) USTSFM. (f) PSSTFN. (g) HS image observed on February 10, 2021 (ground truth).

1) Effectiveness of FEM: We introduce a dilated convolution into FEM, which is conducive to obtaining multiscale receptive fields to enhance semantic-level understanding. To prove the feasibility of FEM, we remove the dilated convolution instead of a single 3×3 convolution layer and keep the rest of settings unchanged [52]. Table IV shows that the multiscale receptive field brings considerable gains to fusion performance.

2) Effectiveness of MAG: We introduce MAG to refine spatial–spectral features and simultaneously suppress redundant information. For comparison, a single 3×3 convolution layer substitute for MAG and the rest of the settings are unchanged. Table IV shows that under all the quantitative metrics, the performance of PSSTFN is better than that of the method without MAG. It is proved that the accuracy of spatial and spectral features fusion can be effectively improved through the guidance of MAG.

3) Effectiveness of FIM: We assume that FIM makes a positive impact on feature adaptive combination, so we eliminated FIM and replaced it with the concatenation operation in the ablation experiment. Table IV demonstrates that FIM delivers more benefits in interlevel feature fusion.

4) Selection of Weight in L_{spe}: For the weight of spectral loss function L_{spe}, the setting of λ_4 can also be a

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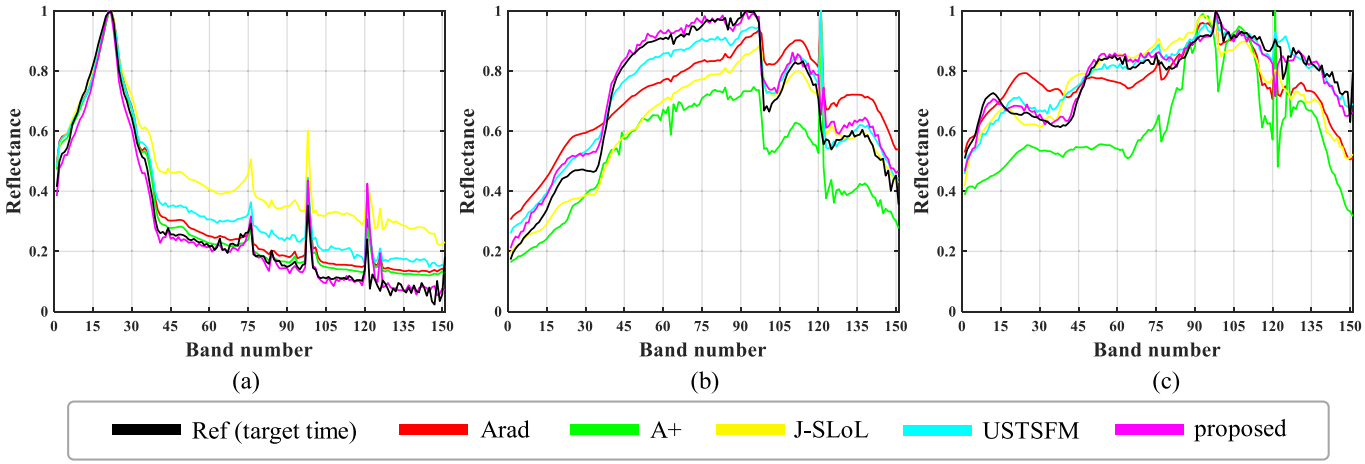


Fig. 7. Spectral reflectance of the comparison methods of typical landscapes on the DY dataset. (a) Waters. (b) Lands. (c) Buildings.

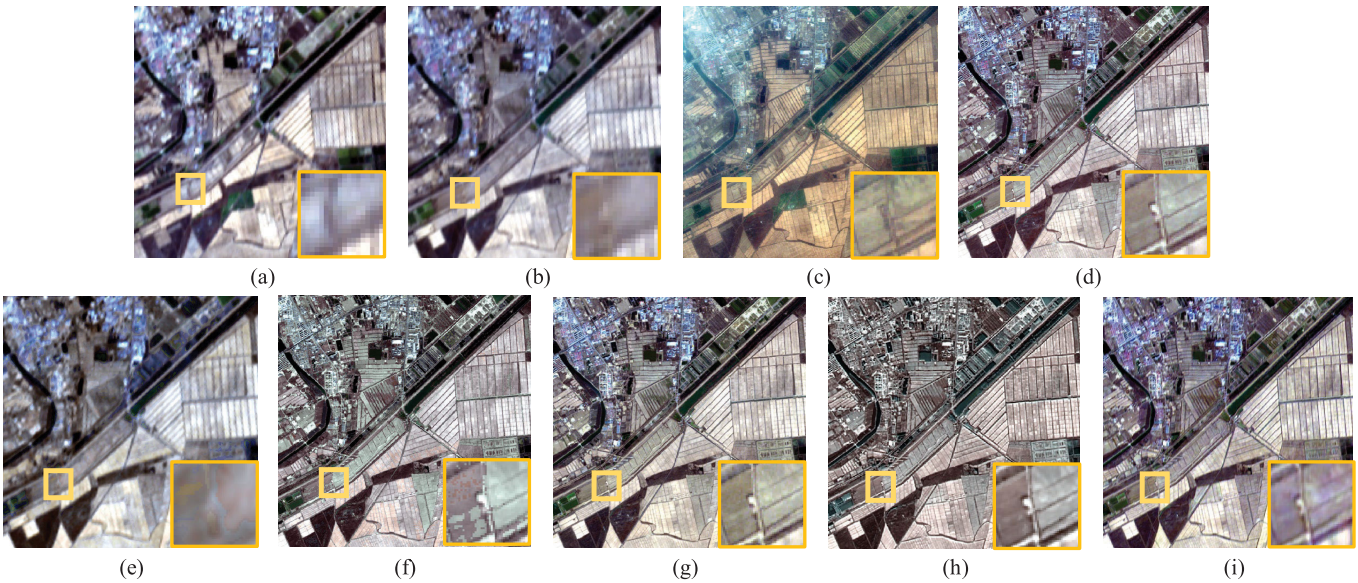


Fig. 8. Full-resolution experimental results on the YRE dataset. (a) HS images observed on December 11, 2019. (b) HS image observed on February 10, 2021 (target time). (c) MS image observed on December 8, 2019. (d) MS image observed on February 10, 2021. (e) Arad. (f) A+. (g) J-SLoL. (h) USTSFM. (i) PSSTFN.

problem worth studying. In the experiments, λ_4 is chosen from $\{1 \times 10^{-1}, 1 \times 10^{-2}, 1 \times 10^{-3}, 1 \times 10^{-4}, 1 \times 10^{-5}\}$. It can be found that PSSTFN achieves the best performance when λ_4 is equal to 1×10^{-2}. In the above-mentioned experiments, λ_4 is determined as 1×10^{-2}. Table V lists the performances of different weights in the SAM loss item.

5) Selection of Loss Function: The loss function L consists of four components L_{MAE1}, L_{MAE2}, L_{spa}, and L_{spe}. It is not enough to provide sufficient constraints for the fusion task by relying solely on L_{MAE2}, and the spatial loss L_{spa} and spectral loss L_{spe} are helpful to reconstruct the edge gradients and spectral structure in training. L_{MAE1} ensures that the spatial–spectral fusion stage of PSSTFN is in effect. Table VI removes L_{spa}, L_{spe}, and L_{MAE1}, respectively, and gets four groups of assessment results, which are based on new loss functions. 1) Without L_{spe}, SAM becomes worse obviously, which confirms that L_{spe} plays a role in spectral fidelity. 2) Without L_{spa}, the decrease in f CC confirms that L_{spa} enhances the ability of spatial detail restoring. 3) Without L_{MAE1}, all the metrics are not satisfactory, which proves that L_{MAE1} is indispensable to the idea of progressive fusion. Table VI verifies the rationality of loss function L.

TABLE IV

ABLATION STUDIES ON THE DY DATASET. (THE BEST IS MARKED IN BOLD AND THE SECOND-BEST IS UNDERLINED)

#	settting	PSNR	ERGAS	SAM	UIQI	CC
1	w/o FEM	29.5832	5.3793	3.9626	0.9297	0.9124
2	w/o MAG	29.6695	<u>5.3553</u>	<u>3.9338</u>	<u>0.9330</u>	<u>0.9139</u>
3	w/o FIM	**30.3109**	5.7165	3.9552	0.9230	0.9045
4	Ours	<u>29.9937</u>	**5.2306**	**3.8021**	**0.9359**	**0.9167**

G. Running Time

As shown in Table VII, we count the testing time of the comparison methods on the DY dataset, and the dimensions of

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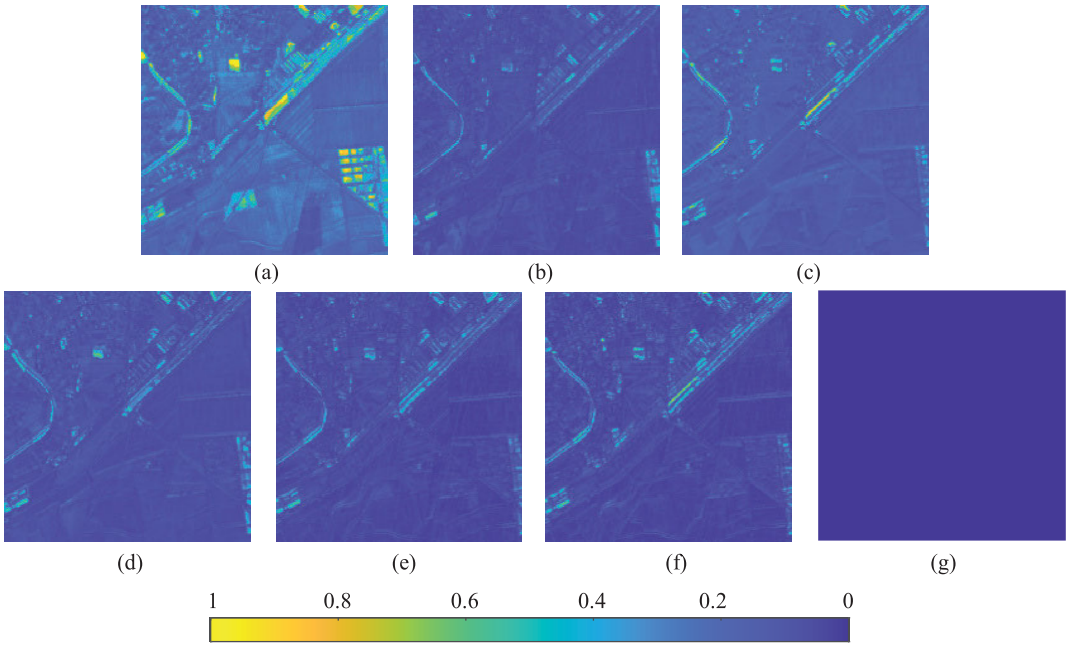


Fig. 9. SAM error maps of full-resolution experiments on the YRE dataset. (a) HS image observed on December 11, 2019. (b) Arad. (c) A+. (d) J-SLoL. (e) USTSFM. (f) PSSTFN. (g) HS image observed on February 10, 2021 (target time).

TABLE V

ABLATION STUDIES ON THE DY DATASET. (THE BEST IS MARKED IN BOLD AND THE SECOND-BEST IS UNDERLINED)

#	settting	PSNR	ERGAS	SAM	UIQI	CC
1	10^{-1}	29.7010	5.7165	3.9552	0.9230	0.9045
2	10^{-2}	**29.9937**	5.2306	**3.8021**	**0.9359**	**0.9167**
3	10^{-3}	29.7854	**5.1535**	3.8760	0.9294	0.9127
4	10^{-4}	29.6610	5.3500	3.9216	0.9305	0.9130
5	10^{-5}	29.6003	5.3116	3.9607	0.9300	0.9108

TABLE VI

ABLATION STUDIES ON THE DY DATASET. (THE BEST IS MARKED IN BOLD AND THE SECOND-BEST IS UNDERLINED)

#	settting	PSNR	ERGAS	SAM	UIQI	CC
1	w/o L_{MAE1}	28.8954	5.7271	4.3024	0.9146	0.8755
2	w/o L_{spe}	29.5281	5.2424	3.9891	0.9238	0.9091
3	w/o L_{spa}	29.1049	5.2881	4.1800	0.9272	0.9001
4	Ours	**29.9937**	**5.2306**	**3.8021**	**0.9359**	**0.9167**

TABLE VII

RUNNING TIME OF COMPARISON METHODS ON DY TEST SET. (THE BEST IS MARKED IN **BOLD**)

Method	Reduced-resolution (s)	Full-resolution (s)	Parameters (M)
Arad	4.800 (CPU)	6.526 (CPU)	-
A+	3.614 (CPU)	4.610 (CPU)	-
J-SLoL	0.495 (CPU)	1.209 (CPU)	-
USTSFM	0.0254 (GPU)	0.0479 (GPU)	7.314
USTSFM	2.2147 (CPU)	16.5285 (CPU)	
PSSTFN	**0.0139** (GPU)	**0.0258** (GPU)	6.469
PSSTFN	1.6227 (CPU)	12.9222 (CPU)	

HR-MS and LR-HS are $600 \times 600 \times 4$ and $200 \times 200 \times 151$, respectively. The traditional methods were implemented on the CPU via MATLAB R2021b, while the DL-based methods were performed on both the CPU and GPU via PyTorch 1.10.0. It can be seen that the DL-based methods usually require greater computational complexity [53], but with the acceleration of GPU hardware, the proposed method still has considerable execution efficiency and is ahead of similar methods.

IV. CONCLUSION

In this article, a progressive STSF network is proposed, which integrates spatial–spectral fusion and spectral–temporal fusion into a unified end-to-end fusion framework. In spatial–spectral fusion, FEM is used to extract features with multiscale receptive fields from HR-MS and LR-HS images, and then MAG serves three feature streams and achieves joint learning of effective content by the mutual guidance of attention mechanism. In spectral–temporal fusion, FIM is developed to merge the hierarchical resulting features with temporal difference, and it plays a role in maintaining spectral structure and eliminating redundancy. We established the DY and YRE datasets for experimental validation of the STSF. Compared with the existing methods, the PSSTFN proves its superiority in quantitative and visual evaluation. In the future, we are committed to developing a robust unsupervised STSF model to solve the degradation in the real-data domain, which requires dealing with thorny spectral constraints and more accurate characterization of temporal changes.

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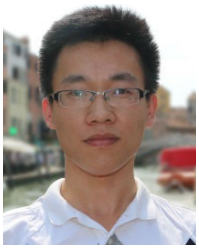
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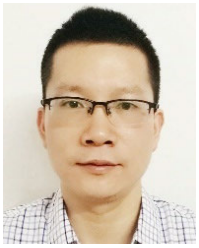
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